

Intelligent Candidate Discovery

Data & AI Challenge

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Problem Statement

Challenge: Data & AI Challenge - Intelligent Candidate Discovery

Goal: Rank 100,000+ candidates for a Senior AI Engineer position.

Constraints:

- High relevance for AI/ML and retrieval engineering skills.
- Zero hallucinations in reasoning.
- Scalable and deterministic.

Approach Overview

We built a 10-Stage Deterministic Scoring Engine:

1. Hard Disqualifiers (Title mismatches, Junior YOE)
2. Lexical TF-IDF Engine
3. Structured AI Skill Score
4. Production & MLOps Score
5. Weighted Fusion via Reciprocal Rank Fusion (RRF)

Data Pipeline

Processing 100k records purely locally:

- CPU-only processing without external LLM dependencies.
- Strict schema validation.
- Memory-efficient JSONL streaming.

Features Engineered

Key Signals Extracted:

- Relevance: FAISS, RAG, Pinecone, BM25, LLM
- Engineering: Kubernetes, Python, Deployment
- Experience: 5-9 Years Sweet Spot Filtering
- Behavioral: Redrob Response Rates

Handling Noise & Honeypots

Explicit filtering ensures quality:

- Managers with no technical AI terms are dropped.
- Non-fit roles (Customer Support, Accountant) are zeroed.
- Junior profiles (<3 YOE) are heavily penalized.

Zero Hallucination Guarantee

Every candidate justification is strictly grounded:

- Reasoning dynamically pulls the ACTUAL title.
- Exact YOE float formatting.
- Verifiable location matching.
- No generated text that cannot be traced to the source JSONL.

Security & Scalability

Designed for Enterprise Trust:

- No external APIs: No data leakage.
- No LLMs: Immune to prompt injection.
- Codebase audited with Bandit and pip-audit.

Top 5 Results Preview

1. CAND_0018499: Senior ML Engineer, 7.2 YOE, Noida
2. CAND_0005260: Senior NLP Engineer, 5.2 YOE, Chennai
3. CAND_0046525: Senior ML Engineer, 6.1 YOE, Pune
4. CAND_0081846: Lead AI Engineer, 6.7 YOE, Jaipur
5. CAND_0042506: Search Engineer, 4.2 YOE, Mumbai

User Interfaces

Included visualization tools:

- Local React Dashboard: For visual review of JSON reports.
- Streamlit Sandbox: Interactive playground testing (sandbox_app.py).

Final Deliverables Summary

Included in the Repository:

- GitHub Repository Source Code
- submissions/submission.csv
- docs/RedrobRank_Official_Submission_Deck.pdf
- sandbox_app.py